

TARUN SINGH

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WORK EXPERIENCE

OMR India Outsources Pvt Ltd

March 2025 – Present

Data Scientist

Agra, India

- Developed a **deep-learning** model to auto-generate multiple-choice questions from video content, driving a **15%** uplift in learner engagement.
- Designed and deployed a personalized **movie-recommendation engine** for the **Reeloid app**, reducing user churn by **12%** and boosting average daily watch time by **35%**.
- Engineered an **AI-powered OMR sheet scanner** to digitize candidate details and answers from any sheet format, cutting processing costs by **45%** and accelerating data capture speed by **55%**.
- Architected and delivered **60+** new RESTful APIs using **FastAPI** and **MongoDB**—while optimizing **100+** existing endpoints—to achieve a **50%** improvement in per-API response times and a **25%** faster overall application performance.

Paytm

June 2023 – February 2024

Quality Assurance Engineer

Bangalore, India

- Automated **100+** test cases using Java and IntelliJ, leading to a **30%** reduction in manual testing efforts and boosting test coverage by **15%**.
- Developed and maintained robust test suites, resulting in a **40%** reduction in production defects and a **25%** increase in product stability.
- Conducted regression testing on code updates and patches to ensure backward compatibility and stability, reducing regression bugs by **30%**.
- Collaborated with cross-functional teams to develop test plans and strategies, resulting in a **29%** reduction in time spent on test case execution and development.

PROJECTS

Bollywood Song Recommendation System | [Website Link](#) | [Source Code](#)

Machine Learning Application

- Developed a recommendation engine by scraping **Spotify API** data and training **ML** models to suggest similar tracks based on user selection.
- Containerized a **Flask** web interface with **Docker** and exposed RESTful endpoints for seamless interaction with the recommendation engine.
- Deployed serverless backend on **AWS Lambda** behind **API Gateway**, ensuring auto-scaling, fault tolerance, and minimal infra management.
- Tech used:** Scikit-learn, Flask, Docker, AWS Lambda, API Gateway, DVC.

Youtube Comments Sentiment Analysis | [Website Link](#) | [Source Code](#)

Deep Learning Application

- Engineered a YouTube sentiment analyzer: fine-tuned **MobileBERT** for 6-emotion classification and used **YouTube API** for comment retrieval.
- Accelerated inference by converting to **TensorFlow Lite** with **Float16 quantization** and serving via a Flask web app containerized in **Docker**.
- Hosted the Flask-based analyzer on **Hugging Face Spaces**, ensuring a scalable, user-friendly experience.
- Tech used:** Transformer, NLP, TensorFlow, Flask, Docker, Hugging Face, DVC.

Harry Potter In Depth Query Bot | [Website Link](#) | [Source Code](#)

Generative AI Application

- Developed a Harry Potter query bot using **LLMs** and **RAG** to deliver precise, context-aware answers about the series.
- Leveraged **LangChain** framework to integrate the LLM, RAG pipeline, and a **Chroma vector database** built from the Harry Potter texts.
- Deployed a **Flask** interface containerized with **Docker** and scaled serverless via **AWS Lambda** behind **API Gateway** for robust execution.
- Tech used:** GPT 4o-mini, LangChain, Chroma, LLM, RAG, Docker, AWS Lambda, API Gateway, Flask, DVC.

TECHNICAL SKILLS

Machine Learning

Deep Learning

Artificial Intelligence

NLP

Data Analysis

Data Science

RAG

Generative AI

NumPy

TensorFlow

Pandas

MongoDB

Statistical Analysis

SQL

LangChain

AWS

Docker

Git

MLOps

Python

FastAPI

INTERNSHIPS AND ACHIEVEMENTS

- Getting Started with Enterprise Data Science, IBM** – Certified by IBM for mastering core data science concepts, tools, and team roles.
- Quality Assurance Engineer Intern, Paytm** – Performed quality assurance testing for web and mobile applications.
- Completed **1000+** data structures and algorithm problems on **GeeksforGeeks** and **Leetcode** combined.

PROFILE LINKS

[Github](#), [Kaggle](#), [Hugging Face](#), [GeeksforGeeks](#), [Leetcode](#)

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University

Bachelor of Technology in Computer Science Engineering

August 2019 - June 2023

Uttar Pradesh, India